
Applied Data Mining

— Week 10—
Spring 2026

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Contents

- Example of K-Means Clustering
- Hierarchical Clustering

K-Means: Example 2

Q: Suppose there are eight points and we need to make three clusters out of them:

A1(2, 10)	A5(7, 5)
A2(2, 5)	A6(6, 4)
A3(8, 4)	A7(1, 2)
A4(5, 8)	A8(4, 9)

- Let us choose three clusters randomly

A1(2, 10)	A4(5, 8)	A7(1, 2)
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- The distance function is given as:

$$\rho(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

Example 2: Cont...

We need to calculate the distance of all points to each of the three centroids and assign each point to nearest centroid to make a cluster.

		(2,10)	(5,8)	(1,2)	
	Points	Dist. Mean 1	Dist. Mean 2	Dist. Mean 3	Clusters
A1	(2, 10)				
A2	(2, 5)				
A3	(8, 4)				
A4	(5, 8)				
A5	(7, 5)				
A6	(6, 4)				
A7	(1, 2)				
A8	(4, 9)				

Example 2: Cont...

For A1:

Point (x1,y1)	Mean1 (x2,y2)	Mean2 (x2,y2)	Mean3 (x2,y2)
A1(2,10)	(2,10)	(5,8)	(1,2)

- Using distance function

$$\varrho(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

A1-Mean 1	A1-Mean 2	A1-Mean 3
$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $
$ 2 - 2 + 10 - 10 $	$ 5 - 2 + 8 - 10 $	$ 1 - 2 + 2 - 10 $
0	5	9

Example 2: Cont...

For A2:

Point (x1,y1)	Mean1 (x2,y2)	Mean2 (x2,y2)	Mean3 (x2,y2)
A2(2,5)	(2,10)	(5,8)	(1,2)

- Using distance function

$$\varrho(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

A1-Mean 1	A1-Mean 2	A1-Mean 3
$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $
$ 2 - 2 + 10 - 5 $	$ 5 - 2 + 8 - 5 $	$ 1 - 2 + 2 - 5 $
5	6	4

Example 2: Cont...

Similarly find the distance of A4, A5, A6, A7 and A8 to all three centroids and keep assigning them to nearest centroid cluster to make a cluster. The table will become:

		(2,10)	(5,8)	(1,2)	
	Points	Dist. Mean 1	Dist. Mean 2	Dist. Mean 3	Clusters
A1	(2, 10)	0	5	9	1
A2	(2, 5)	5	6	4	3
A3	(8, 4)	12	7	9	2
A4	(5, 8)	5	0	10	2
A5	(7, 5)	10	5	9	2
A6	(6, 4)	10	5	7	2
A7	(1, 2)	9	10	0	3
A8	(4, 9)	3	2	10	2

Example 2: Cont...

- Now we get three clusters at three centroids:

Cluster 1

(2, 10)

Cluster 2

(8, 4)

(5, 8)

(7, 5)

(6, 4)

(4, 9)

Cluster 3

(2, 5)

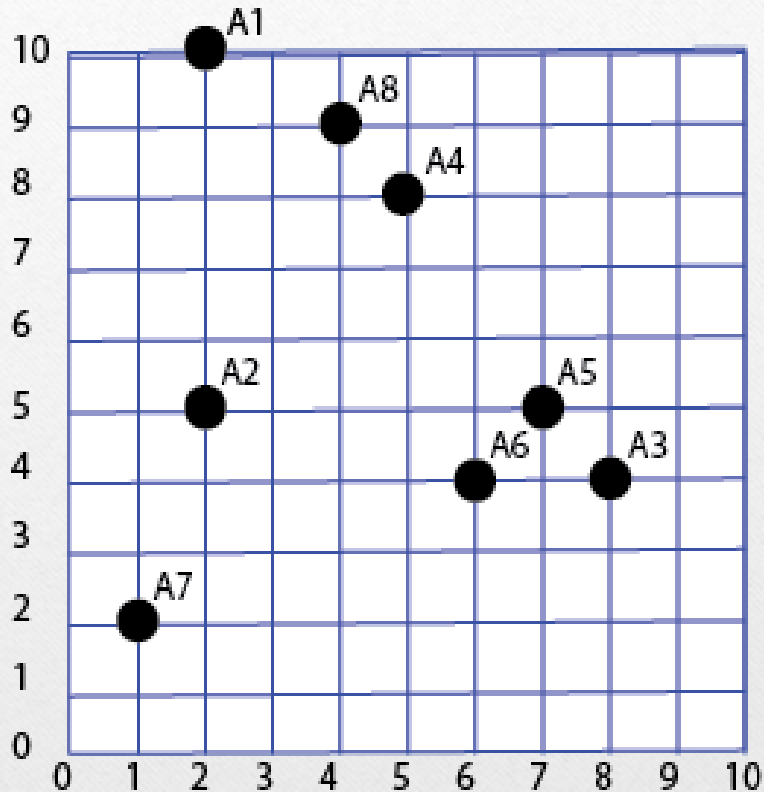
(1, 2)

Cluster 1: Only one point so centroid remains the same.

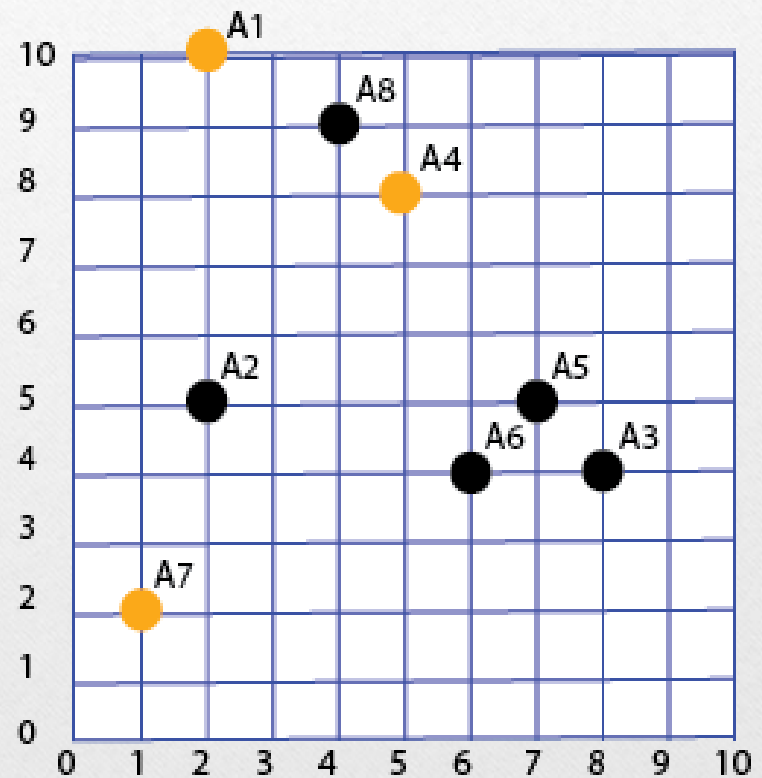
Cluster 2: $((8+5+7+6+4)/5, (4+8+5+4+9)/5) = (6, 6)$

Cluster 3: $((2+1)/2, (5+2)/2) = (1.5, 3.5)$

Example 2: Mapping

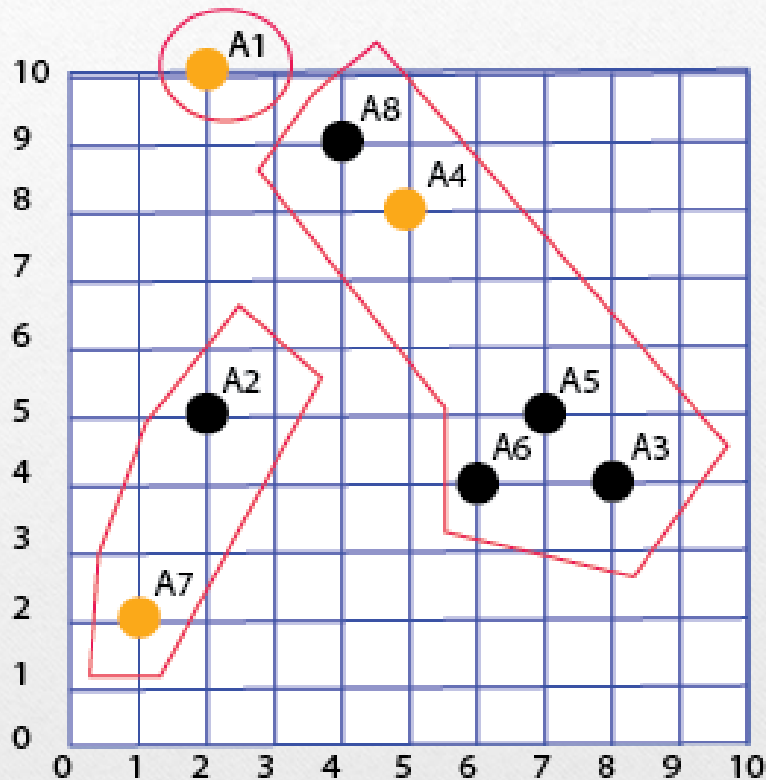


8 Selected Data Points

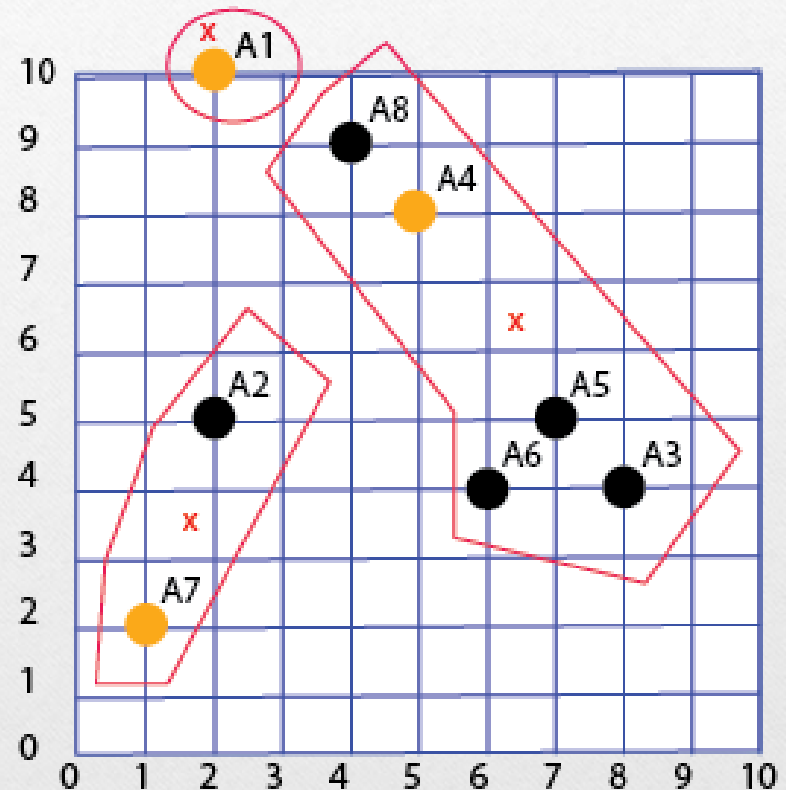


3 initial centroids in
yellow color

Example 2: Mapping



Clusters formed on 3
centroids



Clusters shown in red x

Example 2: Cont...

Iteration 1 is completed.

Now moving onto Iteration 2 using newly computed centroids.

		(2,10)	(6,6)	(1.5,3.5)	
	Points	Dist. Mean 1	Dist. Mean 2	Dist. Mean 3	Clusters
A1	(2, 10)				
A2	(2, 5)				
A3	(8, 4)				
A4	(5, 8)				
A5	(7, 5)				
A6	(6, 4)				
A7	(1, 2)				
A8	(4, 9)				

Example 2: Cont...

For A1:

Point (x1,y1)	Mean1 (x2,y2)	Mean2 (x2,y2)	Mean3 (x2,y2)
A1(2,10)	(2,10)	(6,6)	(1.5,3.5)

- Using distance function

$$\varrho(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

A1-Mean 1	A1-Mean 2	A1-Mean 3
$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $
$ 2 - 2 + 10 - 10 $	$ 6 - 2 + 6 - 10 $	$ 1.5 - 2 + 3.5 - 10 $
0	8	10

Example 2: Cont...

For A2:

Point (x1,y1)	Mean1 (x2,y2)	Mean2 (x2,y2)	Mean3 (x2,y2)
A1(2,5)	(2,10)	(6,6)	(1.5,3.5)

- Using distance function

$$\varrho(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

A1-Mean 1	A1-Mean 2	A1-Mean 3
$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $
$ 2 - 2 + 10 - 5 $	$ 6 - 2 + 6 - 5 $	$ 1.5 - 2 + 3.5 - 5 $
5	5	1

Example 2: Cont...

Similarly find the distance of A3, A4, A5, A6, A7 and A8 to all three centroids and keep assigning them to nearest centroid cluster to make a cluster. The table will become:

		(2,10)	(6,6)	(1.5,3.5)	
	Points	Dist. Mean 1	Dist. Mean 2	Dist. Mean 3	Clusters
A1	(2, 10)	0	8	10	1
A2	(2, 5)	5	5	1	3
A3	(8, 4)	12	4	7	2
A4	(5, 8)	5	3	8	2
A5	(7, 5)	10	2	7	2
A6	(6, 4)	10	2	5	2
A7	(1, 2)	9	9	2	3
A8	(4, 9)	3	5	8	1

Example 2: Cont...

- Now we get three clusters at three centroids:

Cluster 1

(2, 10)

(4, 9)

Cluster 2

(8, 4)

(5, 8)

(7, 5)

(6, 4)

Cluster 3

(2, 5)

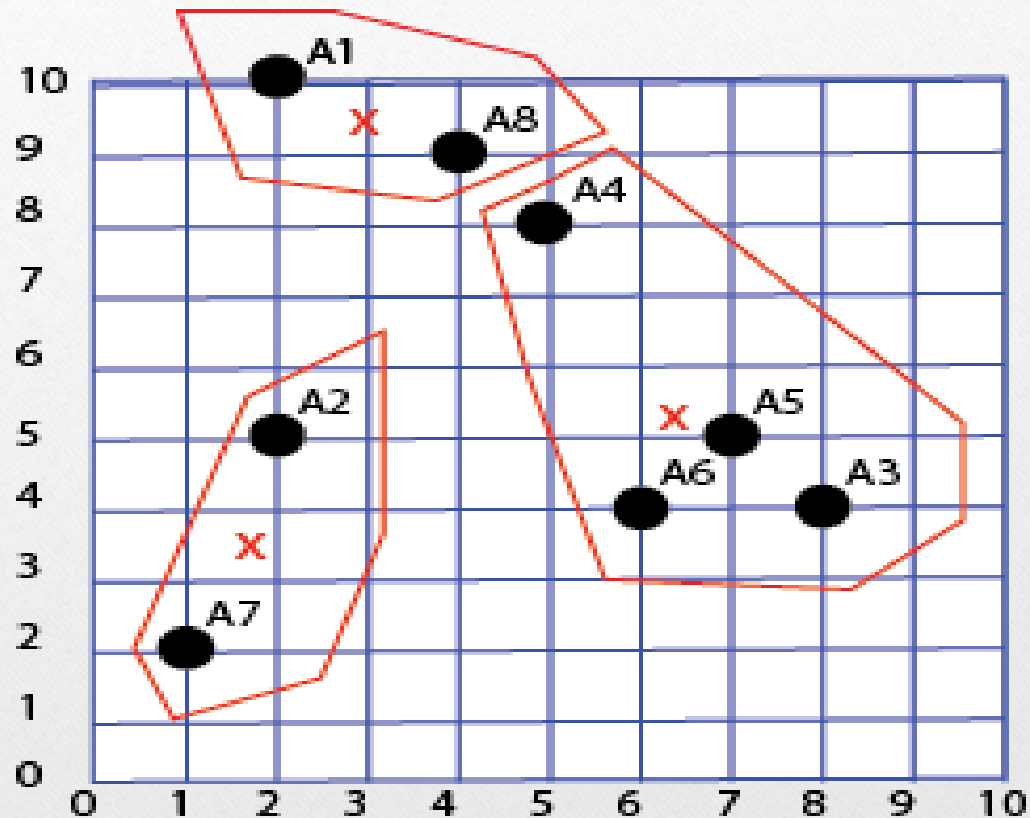
(1, 2)

Cluster 1: $((2+4)/2, (10+9)/2) = \mathbf{(3, 9.5)}$

Cluster 2: $((8+5+7+6)/4, (4+8+5+4)/4) = \mathbf{(6.5, 5.25)}$

Cluster 3: $((2+1)/2, (5+2)/2) = \mathbf{(1.5, 3.5)}$

Example 2: Mapping



New clusters formation after 2nd
iteration

Example 2: Cont...

Iteration 2 is completed.

Now moving onto Iteration 3 using newly computed centroids.

		(3,9.5)	(6.5,5.25)	(1.5,3.5)	
	Points	Dist. Mean 1	Dist. Mean 2	Dist. Mean 3	Clusters
A1	(2, 10)				
A2	(2, 5)				
A3	(8, 4)				
A4	(5, 8)				
A5	(7, 5)				
A6	(6, 4)				
A7	(1, 2)				
A8	(4, 9)				

Example 2: Cont...

For A1:

Point (x1,y1)	Mean1 (x2,y2)	Mean2 (x2,y2)	Mean3 (x2,y2)
A1(2,10)	(3,9.5)	(6.5,5.25)	(1.5,3.5)

- Using distance function

$$\rho(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

A1-Mean 1	A1-Mean 2	A1-Mean 3
$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $	$ x_2 - x_1 + y_2 - y_1 $
$ 3 - 2 + 9.5 - 10 $	$ 6.5 - 2 + 5.25 - 10 $	$ 1.5 - 2 + 3.5 - 10 $
1.5	9.25	7

Example 2: Cont...

For A2:

Point (x1,y1)	Mean1 (x2,y2)	Mean2 (x2,y2)	Mean3 (x2,y2)
A1(2,5)	(3,9.5)	(6.5,5.25)	(1.5,3.5)

- Using distance function

$$\varrho(a, b) = |x2 - x1| + |y2 - y1|$$

A1-Mean 1	A1-Mean 2	A1-Mean 3
$ x2 - x1 + y2 - y1 $	$ x2 - x1 + y2 - y1 $	$ x2 - x1 + y2 - y1 $
$ 3 - 2 + 9.5 - 5 $	$ 6.5 - 2 + 5.25 - 5 $	$ 1.5 - 2 + 3.5 - 5 $
5.5	4.75	2

Example 2: Cont...

Similarly find the distance of A3, A4, A5, A6, A7 and A8 to all three centroids and keep assigning them to nearest centroid cluster to make a cluster. The table will become:

		(3,9.5)	(6.5,5.25)	(1.5,3.5)	
	Points	Dist. Mean 1	Dist. Mean 2	Dist. Mean 3	Clusters
A1	(2, 10)	1.5	9.25	7	1
A2	(2, 5)	5.5	4.75	2	3
A3	(8, 4)	10.5	2.75	7	2
A4	(5, 8)	3.5	4.25	8	1
A5	(7, 5)	8.5	0.75	7	2
A6	(6, 4)	8.5	1.75	5	2
A7	(1, 2)	9.5	8.75	2	3
A8	(4, 9)	1.5	6.25	8	1

Example 2: Cont...

- Now again, we get three clusters at three centroids:

Cluster 1	Cluster 2	Cluster 3
(2, 10)	(8, 4)	(2, 5)
(5, 8)	(7, 5)	(1, 2)
(4, 9)	(6, 4)	

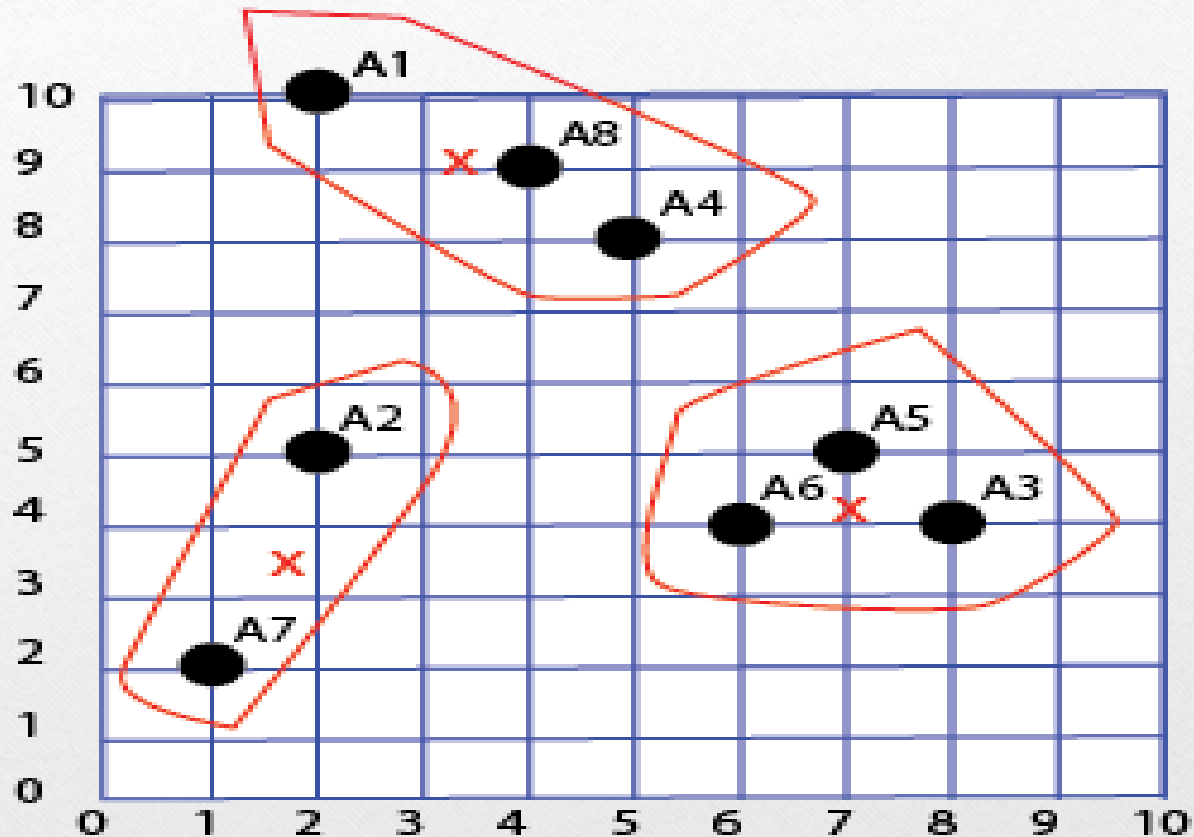
Next, we need to compute the new means (centroids) which is done by taking mean of all points in each cluster.

Cluster 1: $((2+5+4)/3, (10+8+9)/3) = (3.66, 9)$

Cluster 2: $((8+7+6)/3, (4+5+4)/3) = (7, 4.33)$

Cluster 3: $((2+1)/2, (5+2)/2) = (1.5, 3.5)$

Example 2: Mapping



Updated cluster formation after 3rd iteration

Example 2: Cont...

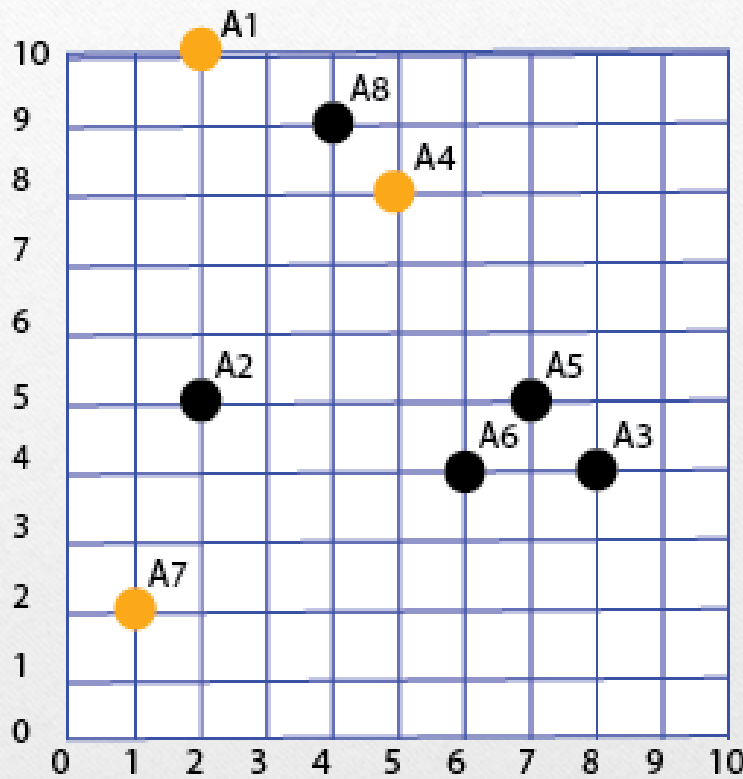
Now if we take the newly computed centroids and compute the values of this table again. It will remain same without any further change and our cluster would become stable.

		(3.66,9)	(7,4.33)	(1.5,3.5)	
	Points	Dist. Mean 1	Dist. Mean 2	Dist. Mean 3	Clusters
A1	(2, 10)	1.5	9.25	7	1
A2	(2, 5)	5.5	4.75	2	3
A3	(8, 4)	10.5	2.75	7	2
A4	(5, 8)	3.5	4.25	8	1
A5	(7, 5)	8.5	0.75	7	2
A6	(6, 4)	8.5	1.75	5	2
A7	(1, 2)	9.5	8.75	2	3
A8	(4, 9)	1.5	6.25	8	1

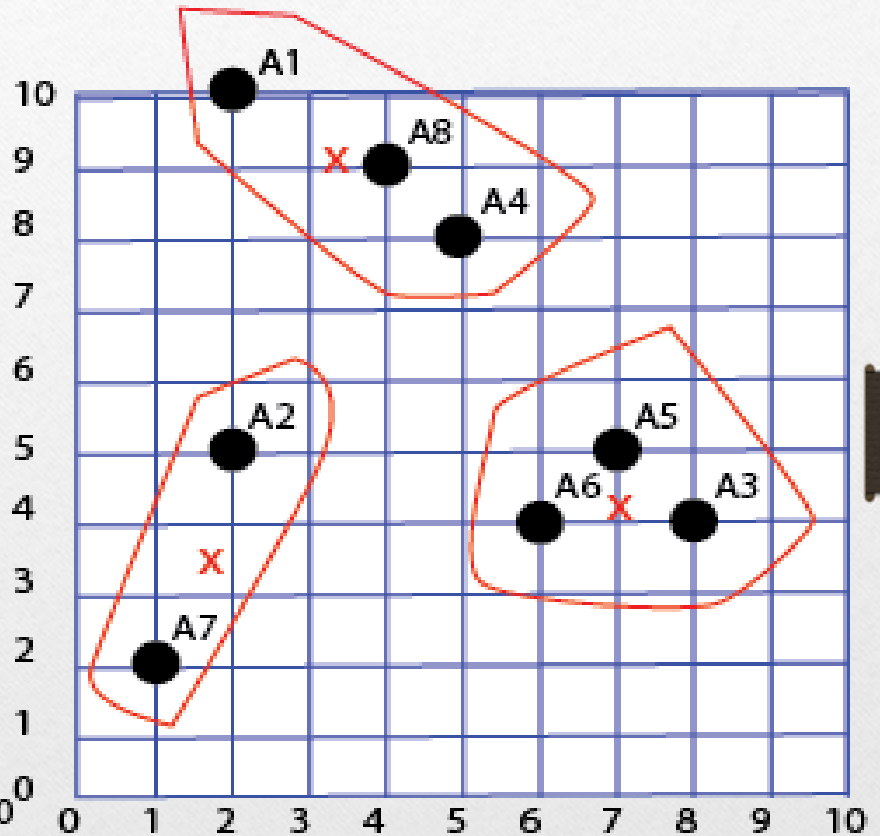
Example 2: Summary

- This cycle will stop after 3 iterations as the values of clusters will not update further.
- The final cluster formation will have 3 clusters at 3 designated centroids.
- This is how the k-means clustering works on unsupervised data using continuous iterations (epochs).

Results Summary



Initial Data with 3
Centroids



Final Cluster formation after 3rd
iteration

Hierarchical Clustering

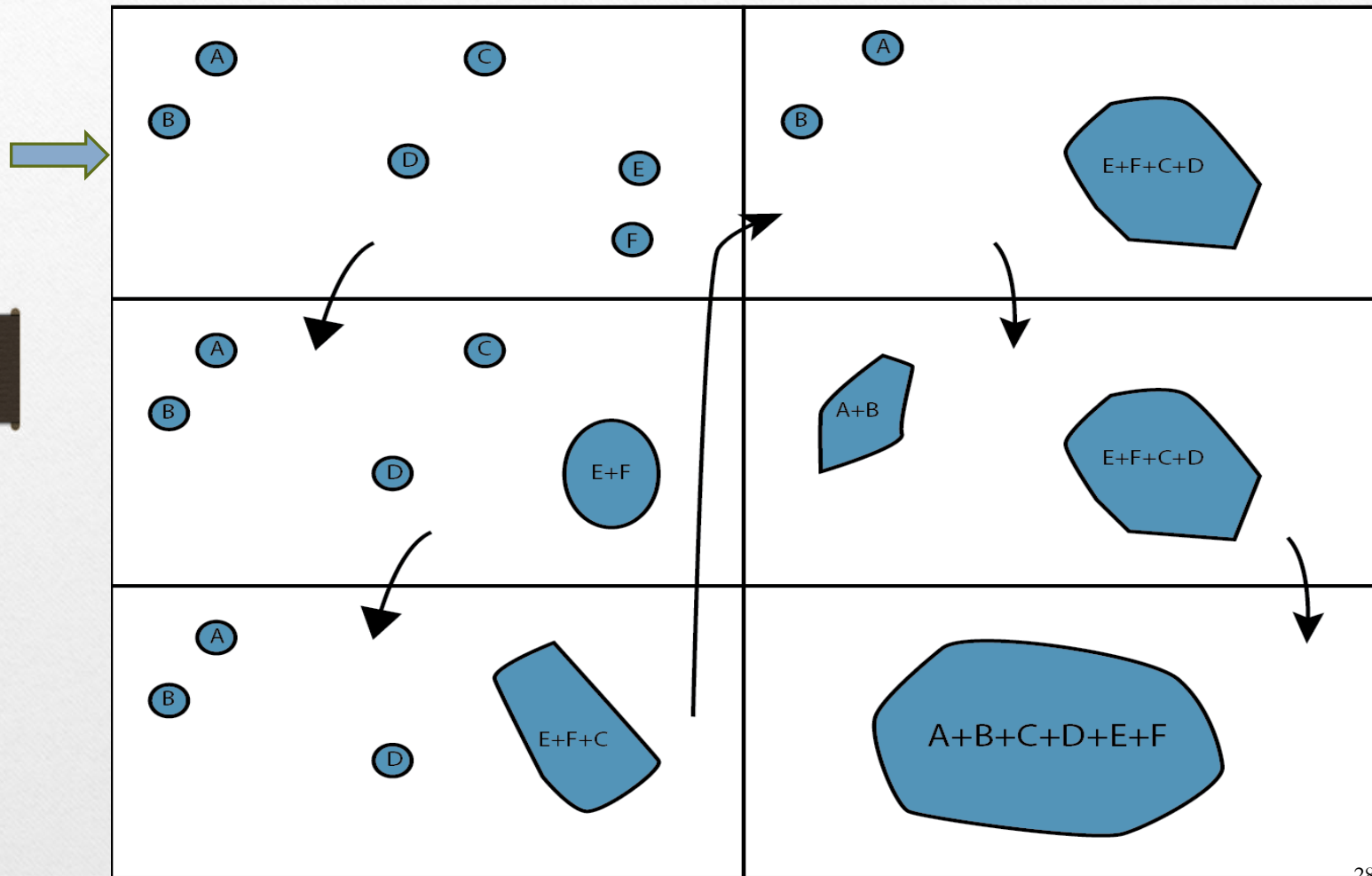
- **Hierarchical clustering**, also known as **hierarchical cluster analysis**, is an algorithm that groups similar objects into groups called **clusters**.
- The endpoint is a set of **clusters**, where each **cluster** is distinct from each other **cluster**, and the objects within each **cluster** are broadly similar to each other.
- The output of hierarchical clustering is a dendrogram which show the hierarchical relationship between the clusters.

Hierarchical Clustering: Steps

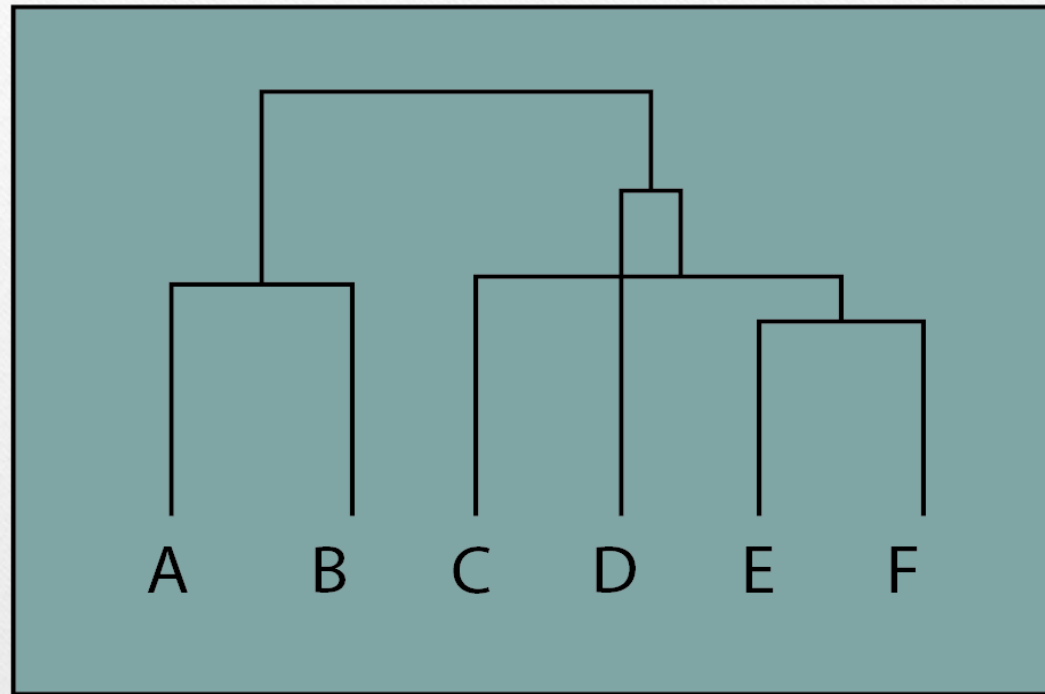
Hierarchical clustering starts by treating each observation as a separate **cluster**.

- Then, it repeatedly executes the following two **steps**:
 1. Identify the two clusters that are closest together, and
 2. Merge the two most similar clusters.
- This iterative process continues until all the clusters are merged together

Hierarchical Clustering: Example



Dendrogram Formation



Dendrogram

Limitations and Advantages

- The dendrogram formations makes it really easy to just understand the relationships at one glance.
- The results are in high correlation with the original data characteristics.
- The focus is on calculating the distances between each cluster rather than centroid.
- There is a lot of detail involved so it can be time complex.

K-Means vs Hierarchical Clustering

K-Means:

- Low memory usage
- Essentially $O(n)$ compute time
- Results are sensitive to random initialization
- Number of clusters is pre-defined

K-Means vs Hierarchical Clustering...

Hierarchical Clustering:

- Deterministic algorithm
- Dendrogram shows us clustering for various choices of K
- Requires only a distance matrix, quantifying how dissimilar observations are from one another
- We can use a dissimilarity measure that gracefully handles categorical variables, missing values, etc
- Memory-heavy, more computationally intensive than K-means

Summary

- Clustering
- K Means Clustering
- Examples

THANK YOU